HEAVY METAL TESTED & CERTIFIED

Infant Formula (Powder) — HMTc Standard

Version 3.0 · 2026-08-05 · native basis: as-sold (powder) · all values µg/kg (ppb)

How every number is set

Each limit is the **strictest maximum level set by any credible government regulator worldwide** — the European Union, United Kingdom, Codex Alimentarius, Health Canada, China, India, Brazil, the Gulf states (GSO), and Australia/New Zealand — converted into the product's own powder, as-sold basis. It is a lookup of sovereign law, not a novel calculation: every value is already in force somewhere, and a certified product sits at or below all of them at once. Where no government regulates the exact product, the limit is read across from the nearest government value for an analogous product. Percentile/occurrence methods are not used; ALARA is reserved for post-publication ratcheting.

Master limit table

Analyte	Tier	HMTc limit	Binding basis
Lead (Pb)	Tier 1	20	strictest government ML (EU 2023/915)
Inorganic arsenic (iAs)	Tier 1	20	strictest government ML (EU 2023/915)
Cadmium (Cd)	Tier 1	10	strictest government ML (EU 2023/915, cow-milk)
Mercury (tHg)	Tier 1	20	government read-across (China GB 2762, canned complementary → formula)
Nickel (Ni)	Tier 2	250	strictest government ML (EU 2023/915)
Aluminium (Al)	Tier 2	3500 cow / 7000 soy	law's own cow/soy split (FSANZ Sch 19)
Tin (Sn)	Tier 2	5000	strictest government ML (India FSSAI, as-sold)
Total chromium (Cr)	Tier 2	2000	strictest government ML (China GB 2762, dairy)

Tier 1 (Pb, Cd, iAs, Hg): zero-tolerance toxics, no safe threshold. Tier 2 (Ni, Al, Sn, Cr): 150% transitional allowance under the Status A–E framework. Total chromium is limited at 2000 (China GB 2762, dairy read-across); hexavalent chromium is controlled through it with mandatory speciation on any total-Cr above 100 ppb, and any quantifiable Cr-VI fails.

Derivation — every government's number, side by side

Cow-milk formula powder. Each row lists every government maximum level in force worldwide, converted to the as-sold powder basis; the HMTc limit is the strictest, and the last column names the binding law. A blank means that government sets no limit for that product; a superscript-[1] value is basis-converted, [2] is a read-across from the nearest government value.

Analyte	EU	UK	CX	CA	CN	IN	BR	GS	AN	HMTc	Binds
Lead (Pb)	**20**	50	70[1]	70[1]	80	200	70[1]	70[1]	70[1]	**20**	EU
Inorg. arsenic	**20**	—	—	—	—	—	140[1]	—	—	**20**	EU
Cadmium (Cd)	**10**	10	—	—	—	100	70[1]	—	—	**10**	EU
Mercury (tHg)	—	—	—	—	—	—	—	—	—	**20**[2]	CN·ra

Analyte	EU	UK	CX	CA	CN	IN	BR	GS	AN	HMTc	Binds
Nickel (Ni)	**250**	—	—	—	—	—	—	—	—	**250**	EU
Aluminium (Al)	—	—	—	—	—	—	—	—	3500[1]	**3500**	AN
Tin (Sn)	—	—	—	—	—	5000	—	—	—	**5000**	IN
Total chromium	—	—	—	—	2000[3]	—	—	—	—	**2000**	CN

All µg/kg, as-sold powder. [1] basis-converted from an as-consumed limit at the 1:7 reconstitution factor. [2] no government sets a mercury ML for infant formula anywhere; nearest government value is China GB 2762 total-mercury 20 for canned complementary infant foods, applied by read-across (FDA: 195 samples, 76% non-detect, max 3.5). [3] total chromium via China GB 2762 milk-powder limit (dairy read-across); Cr-VI has no government limit anywhere, controlled via total-Cr + reflex speciation, any quantifiable Cr-VI fails. Soy powder differs in cadmium (20), nickel (400) and aluminium (7000). CX Codex · CA Canada · CN China · IN India · BR Brazil · GS Gulf/GSO · AN Australia/NZ.

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The literature baseline for every analyte is maintained independently at heavymetalindex.com. This certification standard applies those findings; the two are editorially separate by design.